

Practical week-5

```
requirements.txt
1 fastapi
2 uvicorn
3 sqlalchemy
4 asyncmy
5 pydantic
6 python-jose[cryptography]
7 passlib[bcrypt]
8 python-multipart

PS C:\Users\Himasree\Desktop\college\DEMS\university\library_fastapi_project> python -m pip install -r requirements.txt
Requirement already satisfied: bcrypt>=3.1.0 in c:\users\himasree\appdata\local\programs\python\python311\lib\site-packages (from passlib[bcrypt]) (4.0.1)
[notice] A new release of pip is available: 24.0 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Himasree\Desktop\college\DEMS\university\library_fastapi_project> python -m uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['c:\users\himasree\Desktop\college\DEMS\university\library_fastapi_project']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [900] using StatReload
INFO: Started server process [12056]
INFO: Waiting for application startup.
2026-09-15 10:48:11,874 INFO sqlalchemy.engine.Engine SELECT DATABASE()
2026-09-15 10:48:11,874 INFO sqlalchemy.engine.Engine [raw sql] ()
2026-09-15 10:48:11,875 INFO sqlalchemy.engine.Engine SELECT @@sql_mode
2026-09-15 10:48:11,876 INFO sqlalchemy.engine.Engine [raw sql] ()
2026-09-15 10:48:11,876 INFO sqlalchemy.engine.Engine SELECT @@lower_case_table_names
```

```
library.sql
1 CREATE DATABASE library_db;
2 USE library_db;
3 CREATE TABLE IF NOT EXISTS members (
4     id INT AUTO_INCREMENT PRIMARY KEY,
5     username VARCHAR(100) UNIQUE NOT NULL,
6     password_hash VARCHAR(255) NOT NULL,
7     role VARCHAR(30) NOT NULL DEFAULT 'member'
8 );
9
10 CREATE TABLE IF NOT EXISTS books (
11     id INT AUTO_INCREMENT PRIMARY KEY,
12     title VARCHAR(250) NOT NULL,
13     author VARCHAR(200) NOT NULL,
14     isbn VARCHAR(13) UNIQUE NOT NULL,
15 );
```

#	Time	Action	Message	Duration / Fetch
1	10:31:56	CREATE DATABASE library_db	1 row(s) affected	0.031 sec
2	10:32:02	CREATE TABLE IF NOT EXISTS members (id INT AUTO_INCREMENT PRIMARY KEY, username VARC...	0 row(s) affected	0.078 sec
3	10:32:25	CREATE TABLE IF NOT EXISTS books (id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(250) ...	0 row(s) affected	0.047 sec
4	10:32:50	from sqlalchemy.orm import DeclarativeBase	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL serve...	0.016 sec
5	10:32:56	from sqlalchemy.ext.asyncio import (AsyncSession, async_sessionmaker, create_async_engine)	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL serve...	0.000 sec
6	10:37:26	USE library_db	0 row(s) affected	0.000 sec
7	10:37:29	CREATE TABLE IF NOT EXISTS members (id INT AUTO_INCREMENT PRIMARY KEY, username VARC...	0 row(s) affected	0.062 sec
8	10:37:33	CREATE TABLE IF NOT EXISTS books (id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(250) ...	0 row(s) affected	0.078 sec

[/openapi.json](#)

Authorize

^

[GET](#) / [Home](#)

POST `/register` Register

POST `/login` Login

[POST](#) [/books](#) [Create Book](#)

[GET](#) </books/search> Search Books

POST	/register	Register
------	-----------	----------

Parameters

Cancel

Reset

No parameters

Request body required

```
application/json
```

Edit Value | Schema

```
{
  "username": "divya",
  "password": "library123"
}
```

grant_type

```
string | (string |
null)
pattern: ^password$
```

password

☐ Send empty value

username * required

string

divya

password * required

```
string($password)
```

●●●●●●●●●●

scope

string

scope

☒ Send empty value

Responses

Curl

```
curl -X 'POST' \
  'https://127.0.0.1:8000/login' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/x-www-form-urlencoded' \
  -d 'grant_type=password&username=divya&password=library123&scope=client_id=divya&client_secret=library123'
```

Request URL

Server response

200

Response body

```
{
  "access_token": "eyJ2bG6Gc101ZUzY1N1IsInR5cCI6IHRpKVC3Zy.eyJ2bG6Gc101ZkxZSYsIsInVubGU101ZzZlIjE1X1ILC3JhbnA10ZG3ODh0WTRMDZD.4o3F09957z50H4Zhyv5sXrbye8aucRnj7Fde8Vek",
  "token_type": "bearer"
}
```

Response headers

```
content-length: 186
content-type: application/json
date: Tue, 15 Sep 2026 05:24:05 GMT
server: uvicorn
```

Available authorizations



Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes.

API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Authorized

Token URL: /login

Flow: password

username: divya

password: *****

Client credentials location: basic

client_secret: *****

Logout

Close

POST

/books

Create Book

Parameters

Cancel

Reset

No parameters

Request body

required

application/json

Edit Value

Schema

```
{
  "title": "Python for Beginners",
  "author": "John Smith",
  "isbn": "9781234567890",
  "price": 450.00,
  "published_date": "2024-06-10",
  "description": "An introduction to Python programming."
}
```

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/books' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ2aWQiOiJkaWYyIiwiaXNjaWkiOiJCTlphMDJlZDQ0MTR3MDh9.VzUxXjYyza0QqIzswAjOmng-toeIQ7gs-5YulavQ' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Python for Beginners",
    "author": "John Smith",
    "isbn": "9781234567890",
    "price": 450.00,
    "published_date": "2024-06-10",
    "description": "An introduction to Python programming."
  }'
```

Request URL

http://127.0.0.1:8000/books

Server response

Code

Details

201

Response body

```
{
  "message": "Book added successfully",
  "book_id": 1,
  "added_by": "divya"
}
```

Download

Response headers

```
content-length: 68
content-type: application/json
date: Tue, 15 Sep 2026 05:28:38 GMT
server: uvicorn
```

Responses

GET

/books/search

Search Books

Parameters

Name	Description
q * required string (query)	Python minLength: 1
limit integer (query)	5 maximum: 20 minimum: 1

curl -X 'GET' \

'http://127.0.0.1:8000/books/search?q=Python&limit=5' \

-H 'accept: application/json' \

-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJkaXZSV9IiInJvbnV6bGU1OjE2ZW11ZXI1LCJleHA1OjE3MDh0MTM3MDh9.Vz1KKjXiyza0QgI8zmAjCWng-tocIQ7gs-SYu1avKQ'

Request URL

http://127.0.0.1:8000/books/search?q=Python&limit=5

Server response

Code	Details
200	Response body <pre>{ "query": "Python", "results": [{ "id": 1, "title": "Python for Beginners", "author": "John Smith", "isbn": "9781234567890", "price": 450, "published_date": "2024-06-10", "description": "An introduction to Python programming.", "similarity": 0.03789975769122378 }] }</pre> Response headers <pre>content-length: 247 content-type: application/json date: Tue, 15 Sep 2026 05:33:38 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json